

Théo BEILLE

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Summary

MSc MVA student (ENS Paris-Saclay) and final-year engineering student in applied mathematics at ENSTA Paris (top 5%), applying for the **Student Researcher Program 2027**. Research experience in **learning-to-optimize**: deep learning models that accelerate classical optimisation algorithms while keeping convergence guarantees (journal paper in preparation, target: IEEE).

Research interests: machine learning, deep learning, computer vision, algorithmic foundations of optimisation, inverse problems.

Availability: full-time, 6 months from April 2027 (end-of-studies research internship). Location: Paris, open to other EMEA sites.

Research Experience

Visiting Researcher – KTH Royal Institute of Technology Jan. – Jun. 2026

Supervisor: Prof. Ozan Öktem – Journal paper in preparation (target: IEEE journal) Stockholm, Sweden

- Designed a learned optimisation method: a convolutional neural network (deep unrolling) that predicts the acceleration parameters of an iterative optimisation algorithm, combined with a mathematical safeguard that preserves the algorithm's convergence guarantees.
- Result:** reconstruction up to **10x more accurate** than the classical algorithm after about 10 iterations, evaluated on real clinical CT data (512x512 images).
- Owned the full research loop: literature review, method design, PyTorch implementation, experimental protocol, benchmarking against classical baselines, evaluation metrics and written report.
- Released the code openly: github.com/TheoBeille/Research-Internship. **Stack:** Python, PyTorch, NumPy, Git.

Education

MSc MVA (Mathematics, Vision, Learning) – ENS Paris-Saclay, Université Paris-Saclay 2026 – 2027

Courses: Deep Learning, Computer Vision, Reinforcement Learning, Graphs in Machine Learning, ALTEGRAD (Advanced Learning for Text and Graph Data), Computational Statistics.

ENSTA Paris – Institut Polytechnique de Paris 2024 – 2027

Engineering degree (MSc), Applied Mathematics – Top 5% (216 students), GPA 4.17/4.30 Palaiseau, France

Courses: optimisation, probability, statistics, Bayesian filtering, nonlinear time series, functional analysis.

KTH Royal Institute of Technology – Erasmus Exchange, Applied Mathematics Jan. – Jun. 2026

Courses: nonlinear optimisation, machine learning for stochastic differential equations, stochastic calculus, time series analysis.

Lycée Pierre Corneille – Preparatory Classes (CPGE MPSI / MP*) 2022 – 2024

Intensive two-year programme in mathematics and physics, preparing for the national competitive exams to French engineering schools.

Projects

Time Series Forecasting – ENSTA Paris Python, Statsmodels, SciPy

- Cleaned a real hospital attendance dataset (trend, seasonality, outliers), fitted an ARMA model and produced forecasts with confidence intervals using the innovations algorithm.

Treatment Effect Analysis on Medical Data – ENSTA Paris Python, Pandas, SciPy

- Cleaned a patient dataset and ran a statistical study to measure the effect of a treatment.

Leadership and Teaching

Head of Logistics – Trium ENSTA Paris (career fair) 2025 – 2026

- Organised one of the largest student career fairs in France (160+ companies, 3,000 students, budget over €200,000), leading a team of volunteers from set-up to the event itself.

Mathematics Oral Examiner (kholleur) – CPGE PSI 2026 – present

Weekly oral mathematics examinations for preparatory-class students.

Skills

Programming: Python, C++

Machine learning: deep learning, convolutional neural networks, deep unrolling, learned optimisation, statistical learning, probabilistic modelling, time series

Mathematics: optimisation (nonlinear, iterative methods), inverse problems, probability, statistics, stochastic calculus, functional analysis

Libraries and tools: PyTorch, NumPy, Pandas, SciPy, Statsmodels, Git, GitHub, LaTeX

Languages: French (native), English (C1) | **Interests:** sailing, kitesurfing, swimming, piano